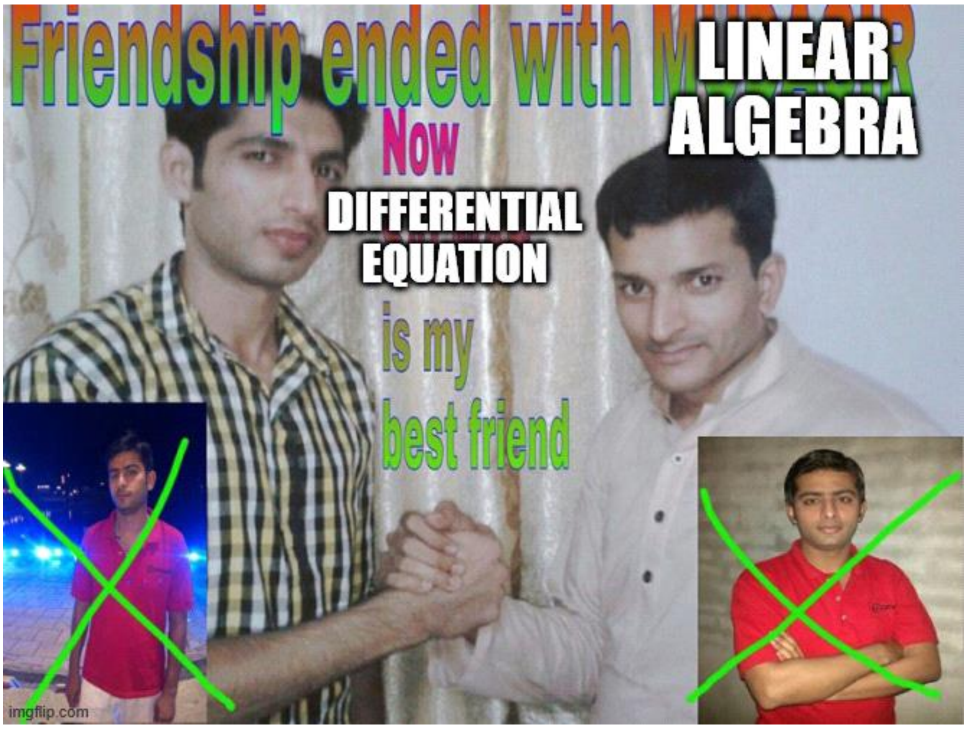
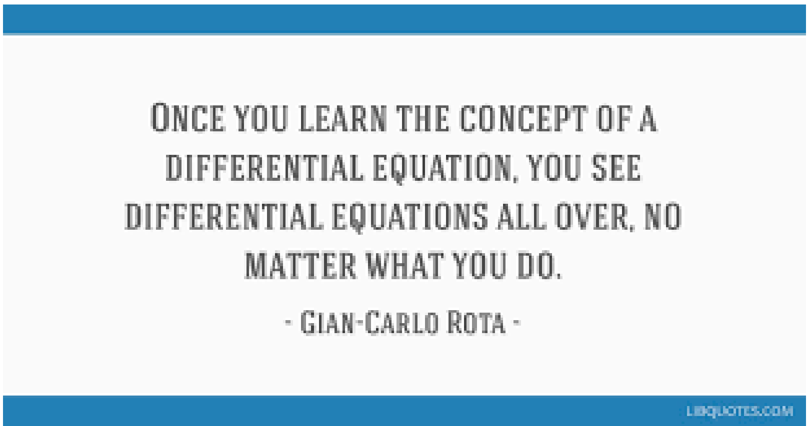
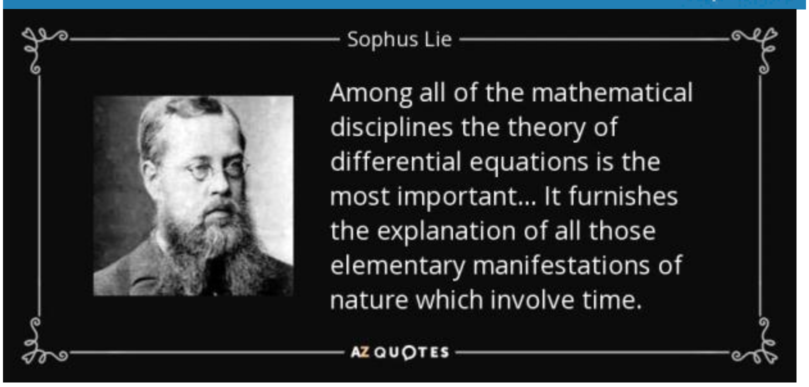
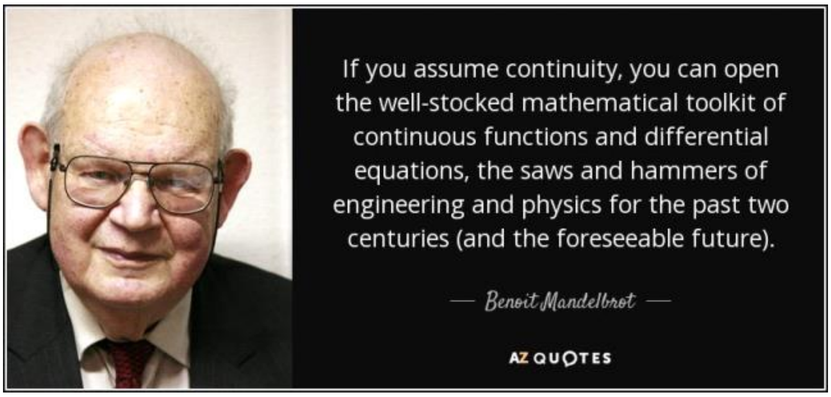
Lect 10.1: Differential Equation - Motivation

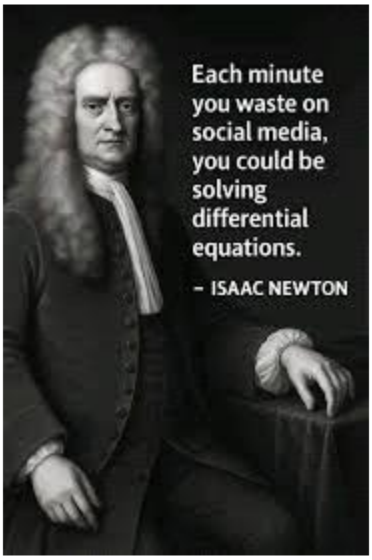
Thursday, 19 March 2026 1:16 pm









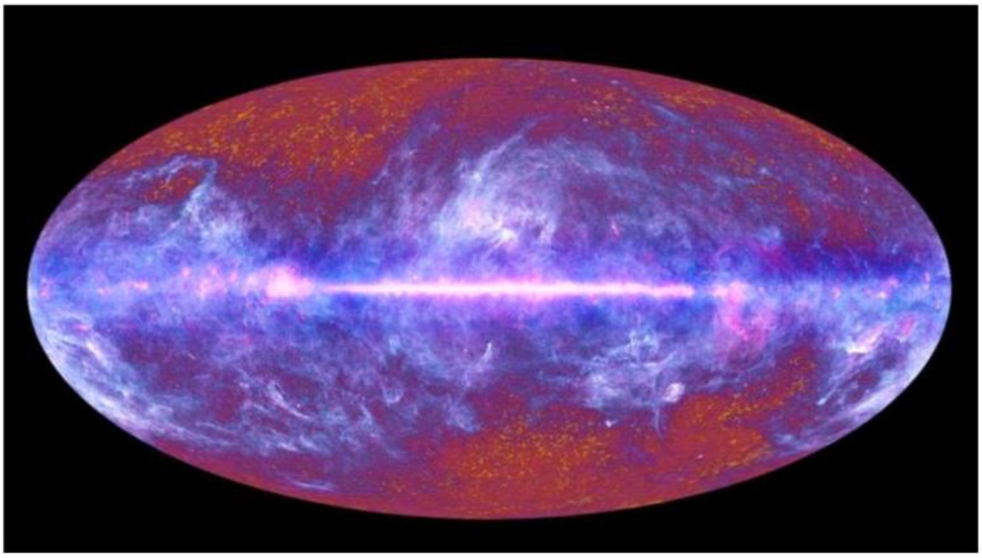


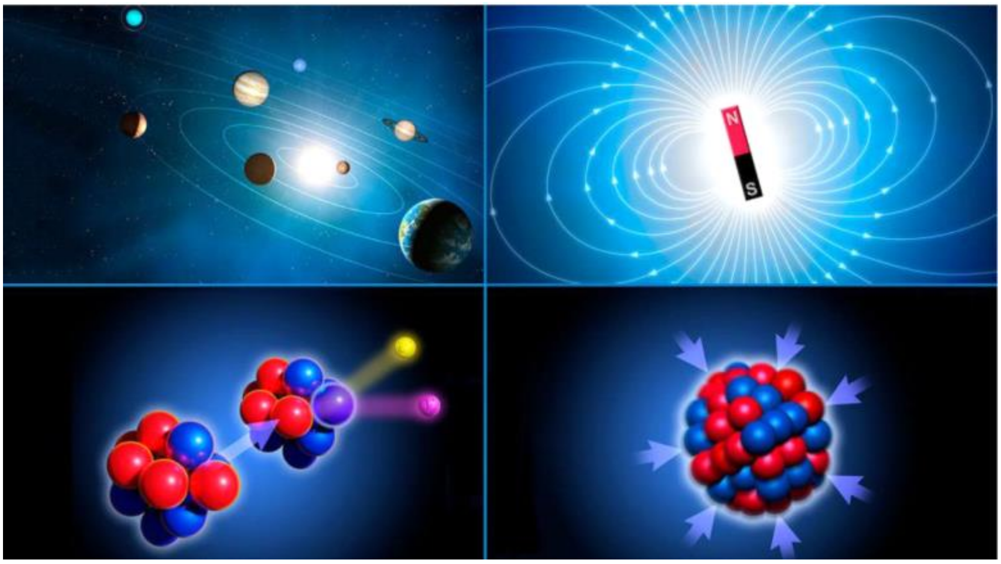
Gottfried Wilhelm Leibniz was the first to state clearly the rules of calculus.

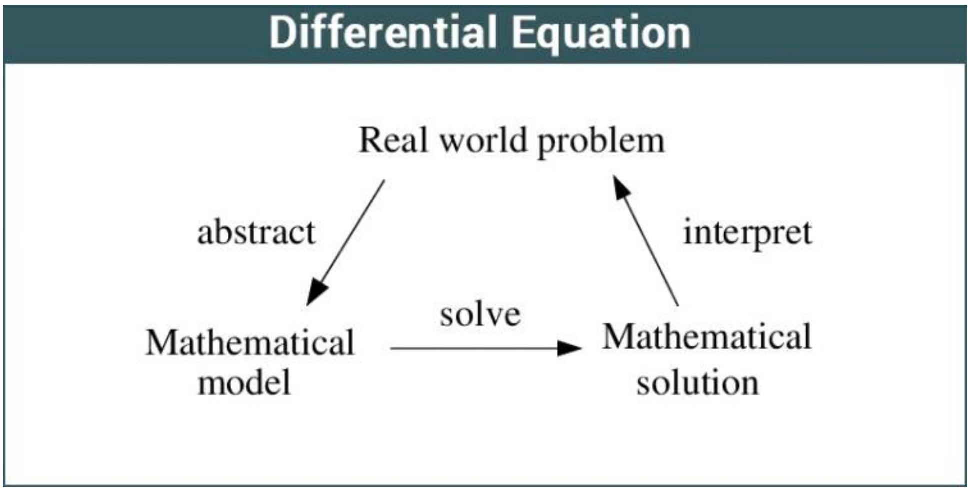
Isaac Newton developed the use of calculus in his laws of motion and universal gravitation.

Why Differential Equations Matters!!!

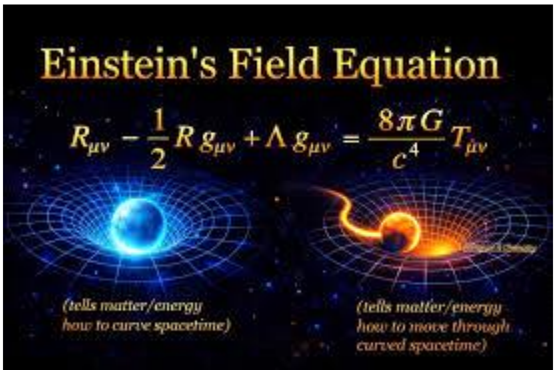
Universe!!!

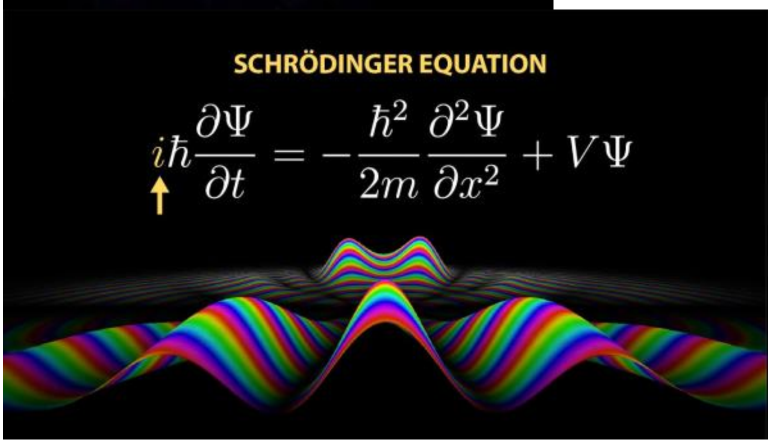


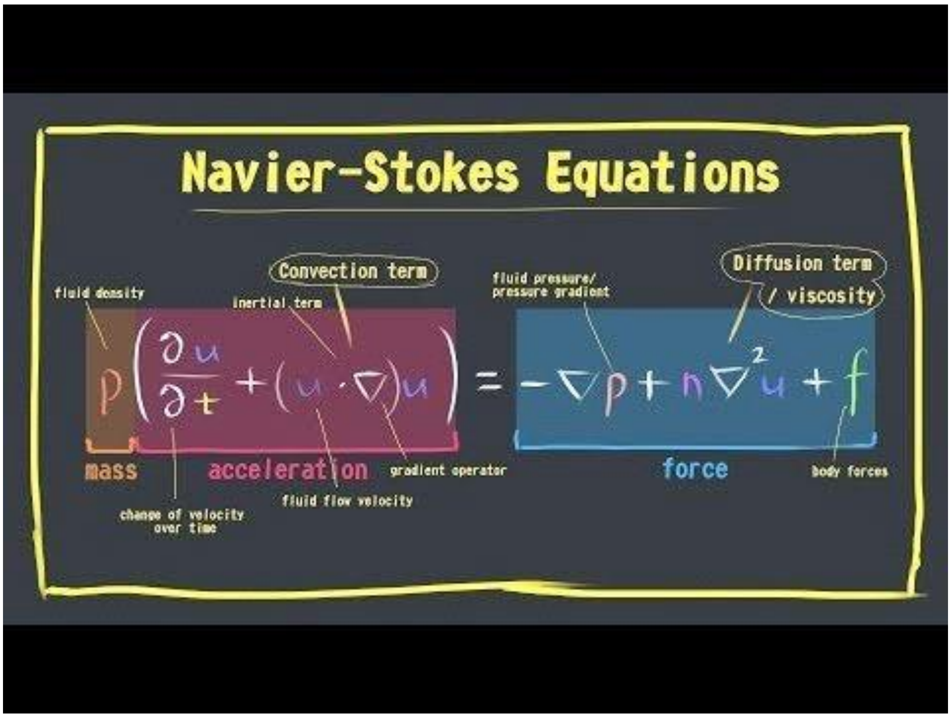


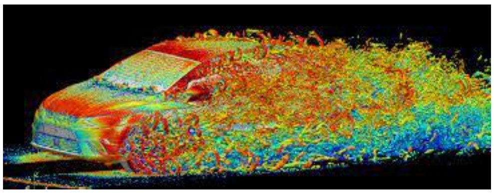


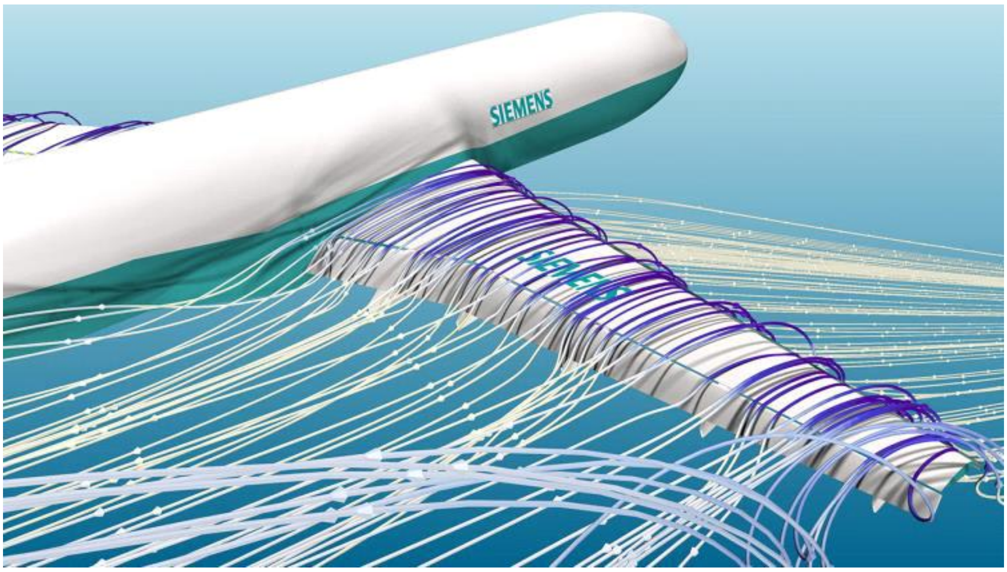
Maxwell's Equations	$\nabla \cdot \mathbf{E} = 0$ $\nabla \times \mathbf{E} = -\frac{1}{c}\frac{\partial \mathbf{H}}{\partial t}$	$\nabla \cdot \mathbf{H} = 0$ $\nabla \times \mathbf{H} = \frac{1}{c}\frac{\partial E}{\partial t}$	J.C. Maxwell, 1865
Second Law of Thermodynamics	$\mathrm{d}S \geq 0$		L. Boltzmann, 1874
Relativity	$E = mc^2$		Einstein, 1905
Schrodinger's Equation	$i\hbar\frac{\partial}{\partial t}\Psi = H\Psi$		E. Schrodinger, 1927
Information Theory	$H = -\sum p(x)\log p(x)$		C. Shannon, 1949
Chaos Theory	$x_{t+1} = kx_t(1 - x_t)$		Robert May, 1975
Black-Scholes Equation	$\frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS\frac{\partial V}{\partial S} + \frac{\partial V}{\partial t} - rV = 0$		F. Black, M. Scholes, 1990

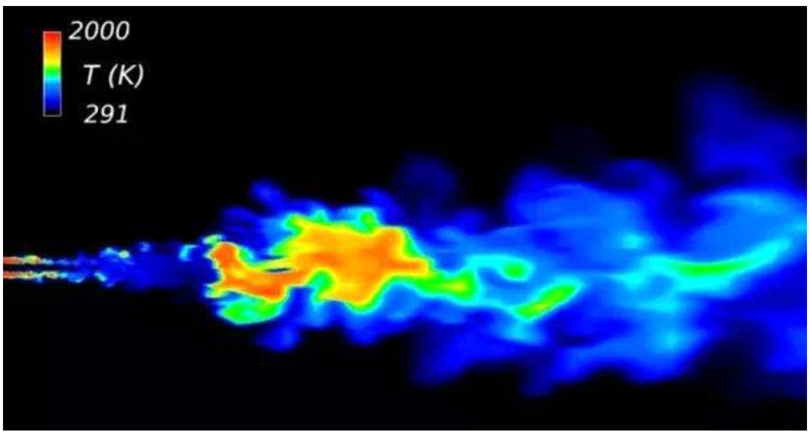





Navier-Stokes Equations
Convection term
fluid density
inertial term
fluid pressure/
pressure gradient
Diffusion term
/ viscosity
$\rho\left(\frac{\partial u}{\partial t}+(u\cdot\nabla)u\right)=-\nabla p+n\nabla^2 u+f$
mass
acceleration
gradient operator
force
body forces
fluid flow velocity
change of velocity
over time




SIEMENS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The Navier–Stokes existence and smoothness problem is a $1 million Millennium Prize challenge from the Clay Mathematics Institute. It asks to prove whether smooth, globally defined, three-dimensional solutions exist for the Navier–Stokes equations (which model fluid flow) or if they break down, failing to describe turbulent behavior at high speeds. Clay Mathematics Institute +2